Scotty: A Compact System-1 Decision Model with Balanced Beam Supervision

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Abstract. Many software tasks need a choice rather than a written answer. We present Scotty, a compact System-1 model that reads a situation, a question and candidate answers, then scores all candidates in one forward pass. A pretrained ModernBERT encoder supplies language understanding; a newly initialized decision head turns its representations into one score per option. Training uses Beam-generated decisions together with Typed Decisions, sampling the two sources equally in every effective batch. Across three training seeds, the model obtains a mean validation objective of 0.859532 with sample standard deviation 0.004603. On a group-safe source audit, the seed-0 model reaches 0.841031, including 76.67% Typed accuracy and 88.62% Beam raw accuracy. The audit is drawn from the same source collections as training, so it is not an external or fresh-distribution test. Beam data generation stayed within the $25 API-credit budget supplied to challenge participants; this excludes local training and hardware costs. Scotty's softmax values are comparative scores, not calibrated confidence, and the model does not generate reasoning.

1 Introduction

A parcel-release service, a support router and a maintenance dashboard may look unrelated, yet each repeatedly asks the same kind of question: given the facts, which answer from a known list should be selected? A general text generator can answer such questions, but it also produces words that the application must parse and check. A dedicated decision model can instead return scores for the allowed answers directly.

Scottyis built for this narrower interface. It is a 421-million-parameter model that selects among options supplied with each request. It handles yes/no decisions, named categories and ordered ratings with the same input format. We call the path *System-1* in an engineering sense: the model produces a direct answer in one forward pass, without a generated chain of thought. The name does not imply human cognition, guaranteed correctness or reliable intuition outside the evaluated setting.

The central design is simple. Reflection Beam prepares varied practice decisions with target answers [1]. The public Typed Decisions collection contributes a second source of structured examples [2]. A pretrained ModernBERT encoder [3] is then fine-tuned with a fresh Laya-format decision head [4]. Beam transfers knowledge through supervised examples, a form of distillation [5]; no Beam weights are copied into Scotty.

2 A decision, step by step

Every request has three main parts. The *situation* supplies relevant facts, the *question* says what must be decided, and the *options* define the answers the application can use. Making these parts explicit keeps the output constrained and inspectable.

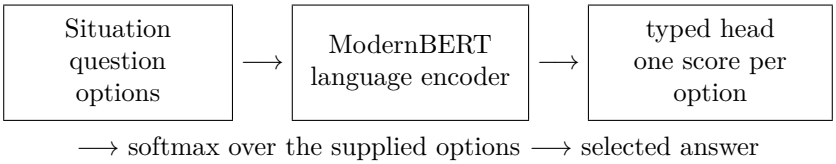
Situation: A parcel is sealed and labelled, but its final weight has not been recorded. The release rule requires all three checks.

Question: Is the parcel ready for release?

Options: No; yes.

The rule supports **no** because one required check is missing. This example explains the interface; it is not a measured model prediction.

The request is serialized as one sequence. The question and every candidate are placed before the situation, and each candidate receives a special `[MASK]` marker. ModernBERT reads the complete sequence jointly, so the representation of each option can depend on the situation, question and competing answers.



The fresh decision head adds a learned embedding for the three decision types, applies two Transformer layers and uses one shared scalar scorer at the option markers. Softmax converts the resulting logits into non-negative values that sum to one. The largest value selects the answer. These values describe relative preference among the supplied options; without separate calibration they should not be read as a verified probability that the answer is correct.

Candidates are scored together, not in independent chunks. The full input is limited to 1,024 tokens. The question and option region has a 256-token budget, and each option is capped at 48 tokens. If important facts fall outside the retained context, the model cannot use them. No rationale or free-form explanation is generated on this path.

3 Training Scotty

3.1 Two complementary sources

Beam was used as a teacher that writes practice decisions and their target answers. After filtering, the Beam collection contains 19,878 decisions across 18 operational task families. Examples cover prerequisite checks, routing choices and ordered assessments. Automated checks reject malformed examples, invalid targets and detected duplicates; similarity screening and group-aware splitting reduce obvious leakage. They cannot guarantee that every teacher label is correct or every remaining example is semantically unique.

The API calls used to generate the Beam dataset stayed within the $25 credit budget provided to challenge participants. This statement covers teacher generation only. It does not include the cost of local training, hardware or later evaluation.

Typed Decisions provides structured cases from workflows such as customer service, invoices, security and agent traces [2]. Its targets can express a distribution across answers rather than only one winner. For example, a teacher could place 0.7 on one rating and 0.3 on its neighbour. Beam examples use a single target option. Cross-entropy training supports both formats: hard Beam targets become one-hot distributions, while Typed targets retain their supplied weights.

Source	Training decisions	Validation decisions
Beam-generated	13,292	3,327
Typed Decisions	4,800	1,200
Available total	18,092	4,527

Table 1: Training and validation source pools. Sampling controls how often each source is presented; it does not change these unique-example counts.

3.2 Balanced presentation

Scotty samples Beam and Typed examples equally: half of each effective batch comes uniformly from Beam and half uniformly from Typed. This prevents the larger Beam pool from automatically dominating the number of presentations while retaining diversity within each source. It is source balance, not class balance, and it does not imply equal token counts or equal compute per source.

Training starts from a pretrained `answerdotai/ModernBERT-large` encoder and a newly initialized decision head. The entire encoder and head are updated. The reported model uses seed 0, 849 optimizer updates, effective batches of 64 and 54,336 total presentations. Beam hard targets and Typed soft targets share the same cross-entropy objective. Its checkpoint has SHA-256 `e79b758aea590bcc4504e6b622c206080a7d1fedc49d572d3a185321b8e2eeda`.

4 Evaluation

The evaluation asks two different questions. Typed hard accuracy measures how often the model's top option matches the target's top option. Beam raw accuracy measures the same quantity over Beam decisions. Beam position-balanced accuracy first forms observed groups by decision type, number of options and target position, then gives those groups equal weight. It reveals uneven behaviour that a frequent group could hide in a raw average.

The primary validation objective is

$$S = \frac{1}{2}(\text{Typed hard accuracy}) + \frac{1}{2}(\text{Beam position-balanced accuracy}).$$

Three independent Scotty runs obtain mean $S = 0.859532$ with sample standard deviation 0.004603. The corresponding mean Typed hard accuracy is 0.798611, and mean Beam position-balanced accuracy is 0.920454. The standard deviation summarizes three observed seeds; it is not a population guarantee or a confidence interval.

The source audit contains 1,344 Beam decisions and 480 Typed decisions from 96 Typed cases. Related examples remain grouped, and these audit examples are excluded from training and validation.

Source-audit metric	Scotty
Primary objective S	0.841031
Typed hard accuracy	0.766667 (368/480)
Beam raw accuracy	0.886161 (1,191/1,344)
Beam position-balanced accuracy	0.915395

Table 2: Seed-0 results on the group-safe source audit.

The audit evaluates withheld groups from the same source collections used for training. It is therefore not a fresh distribution or independent external benchmark. The objective combines two metrics and should be read together with the separate Typed, Beam raw and Beam position-balanced values in Table 2. No case-grouped confidence interval can be reconstructed because the audit retained aggregates rather than per-example predictions.

These limits matter. The results describe Scotty on the two known sources; they do not establish universal generalization, freedom from contamination or readiness for high-stakes autonomous decisions.

5 Using Scotty in an application

At inference time, an application supplies a situation, a question and at least two distinct options. Scottyjointly encodes them, returns one score per option and chooses the largest. The interface is especially useful when the downstream system already knows the valid actions: route to one team, approve or reject a step, or select one level on a scale.

Decision type	Example question	Supplied options
Boolean	Are all release checks complete?	No; yes
Category	Where should this request go?	Repairs; billing; logistics
Ordered	How urgent is this issue?	Low; medium; high; critical

Table 3: One candidate-scoring interface supports several application shapes.

The model is attractive when constrained outputs simplify integration: no answer parser is needed, invalid labels cannot be generated, and the application can log every candidate score. Those benefits do not remove the need for careful question design. Vague instructions, missing facts, overlapping options and long inputs can all produce misleading results.

Softmax scores should be treated as rankings unless calibration has been measured for the deployment domain [6]. A value of 0.9 is not automatically a 90% chance of correctness. A safe workflow can set thresholds only after calibration, record the input and option set, and escalate unfamiliar or consequential cases to a larger model or a person. Because Scottydoes not generate reasoning, applications that require an explanation must obtain it separately rather than presenting the score as a rationale.

5.1 Limitations

Both principal sources use synthetic supervision, so teacher errors and shared biases can be learned. Filtering was assistant-assisted rather than an independent expert annotation study. Three training seeds are a small sample, and the audit is drawn from known source collections. No real-world deployment or fresh-distribution evaluation is reported. The combined objective and every component metric should be examined separately because an aggregate can hide uneven behaviour across groups.

The 421-million-parameter model is compact relative to its teacher, not universally small. Memory, latency and throughput depend on hardware, precision, input length and batching. We therefore make no broad speed claim here. Data and model licenses, privacy requirements and domain-specific harms must be reviewed before deployment.

6 Conclusion

Scottyturns teacher-generated practice decisions into a reusable local decision component. The pretrained ModernBERT encoder supplies language representations; the newly initialized typed head scores the answers that the application provides. Training presents Beam and Typed Decisions equally, so both sources contribute half of each effective batch. The configuration is stated exactly and the model is identified by its checkpoint hash.

The measured results are promising but deliberately bounded. Across three seeds, Scotty reaches validation $S = 0.859532 \pm 0.004603$ (mean and sample standard deviation). The source audit records $S = 0.841031$, Typed hard accuracy of 76.67%, Beam raw accuracy of 88.62% and Beam position-balanced accuracy of 91.54%. Reporting the component metrics alongside S makes the system's behaviour more visible than a single aggregate alone.

Future evaluation should use a fresh distribution, preserve per-example predictions, include more training seeds and measure calibration explicitly. Within its present evidence boundary, Scottyoffers a clear pattern for applications that need a fast, constrained choice rather than generated prose: spend a larger teacher's effort on supervision, train a smaller scorer, and keep the options and uncertainty visible to the surrounding system.

References

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